



Synergistic toughening effects of grafting modification and elastomer-olefin block copolymer addition on the fracture resistance of wood particle/polypropylene/elastomer composites

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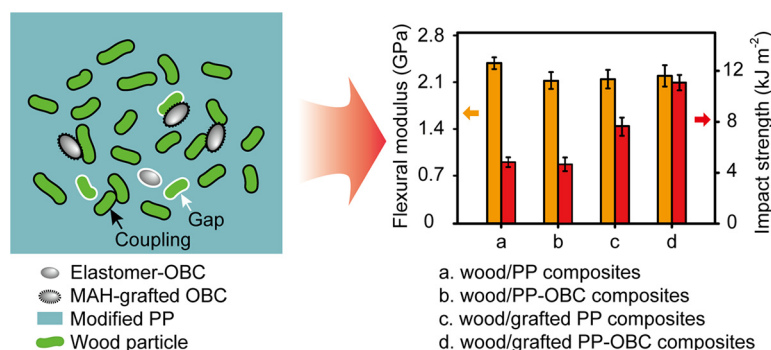
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HIGHLIGHTS

- The toughening effect of OBC on wood particle-filled PP composites was better than that of the commercial elastomer (POE).
- Matrix modification with MAH was an effective method to enhance interfacial interactions among PP, OBC, and wood particles
- By adding OBC and undertaking grafting modification, the impact strength was increased, without compromising stiffness.

GRAPHICAL ABSTRACT



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ABSTRACT

A novel wood particle/polypropylene/elastomer composites with good impact resistance and stiffness was prepared by adding elastomer-olefin block copolymer (OBC) and grafting maleic anhydride (MAH) on polypropylene/OBC blends. The grafted MAH greatly improved the interfacial interaction among polypropylene, OBC, and wood particle, resulting in an enhanced impact resistance, stiffness and dimensional stability of the resulting composites. The modified matrix and their composites were characterized by Fourier transform infrared spectroscopy, scanning electron microscopy, mechanical tests, and dynamic mechanical analysis. The results confirmed that with 60 wt% wood particles loading, the synergistic interactions between OBC blending and MAH grafting improved the impact strength of the MAH-grafted blends/wood particle composites by 49%, with stiffness comparable that of the MAH-grafted polypropylene/wood particle composites. The composites modified with MAH also revealed an enhanced interfacial adhesion among polypropylene, OBC, and wood particles, as evidenced by dimensional stability tests, dynamic mechanical analysis, and morphology observation.

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1. Introduction

The production and application of wood plastic composites (WPCs) has recently undergone rapid and steady development due to economic

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concerns and the environmental properties of lignocellulosic particles (LPs) [1,2]. WPCs are most used in building, construction, furniture, and automotive industries. However, in structural and engineering applications, WPCs must meet the requirements of high stiffness and toughness simultaneously, especially for commercial polypropylene (PP)-based WPCs with high LP loading (up to 70 wt%) [3]. An additional complication is that relatively hydrophilic LP is not compatible with the hydrophobic polyolefin matrix. This incompatibility results in insufficient wetting of the LP by the matrix and LP agglomeration during manufacturing, with low fracture resistance of the resulting WPCs [4].

The use of coupling agents/compatibilizers is a rapid, effective strategy to increase interfacial bonding between the LP and the matrix [5]. Maleated polyolefins are the most commonly used coupling agents, and can improve the mechanical properties of the composites, especially for tensile and flexural strength [6]. Another strategy is grafting of the polymer matrix with polar monomers to improve filler dispersibility and interfacial bonding [7,8]. These strategies can also improve the tensile and flexural properties of the composites, but there is little positive influence on the impact toughness of the composites for high value applications.

To more substantially improve the impact toughness of polyolefin-based WPCs, elastomers have been used together with maleated polyolefin [9–11]. In one study, an elastomer was used to improve the impact resistance and maleated polyolefin was used to offset the adverse effects of the elastomer on the strength and modulus of resulting wood-based composites by improving interfacial adhesion [12]. In this kind of three-component system, two intermediate structures may be formed: (1) separate dispersions of elastomers and rigid reinforcements in the polymer matrix [13], and (2) encapsulation of rigid reinforcements by elastomers to form an embedded structure [14]. However, in practical use, the elastomer may form a third separate phase in the matrix rather than an encapsulating interface due to incompatibility, which may reduce the toughening effect of elastomer addition [5]. These embedded structures in the composites can cause a slight increase in the impact strength and modulus [14]. Sudarel et al. found that the addition of maleated ethylene-propylene-diene (EPDM) copolymer prompted the formation of an embedded structure [15]. However, this positive effect of an embedded structure in PP/wood/elastomer composites was offset by the adverse softening effect of low-modulus elastomer at high content, resulting in decreased stiffness and strength [16]. To address this, Sudar et al. added both elastomers and stiff fillers to WPCs to simultaneously increase toughness and stiffness [16]. Such stronger reinforcements and elastomers can be used to compensate for each other's shortcomings in WPCs to provide significant toughness and stiffness [12]. However, the addition of mineral filler or glass fiber may result in severe abrasion of the extruder screw during processing [1].

A novel multi-block thermoplastic elastomer, olefin block copolymer (OBC), was introduced by the Dow Chemical Company in 2006 and received considerable attention. This multi-block copolymer comprises crystallizable ethylene-octane blocks (hard blocks) with very low co-monomer content and amorphous ethylene-octane blocks (soft blocks) with high co-monomer content [17]. Compared with traditional elastomers (such as polyolefin elastomer and EPDM), OBC-toughened composites showed a good rigidity-toughness balance [11,18,19]. In these studies, the combination of rigid reinforcement at a low concentration (less than 40 wt%) and maleated elastomer was an effective approach to simultaneously achieve large impact resistance and stiffness simultaneously [20,21]. However, at high filler loading levels, it is unclear how the compatibility of each component affects the fracture behavior of PP/elastomer/filler composites.

In our previous work [7,23], grafting of polymer matrix with maleic anhydride (MAH) was shown to enhance the interfacial bonding and improve the dispersion of wood particles in the modified matrix. Better mechanical properties of MAH-grafted PP/wood particle composites were also obtained relative to those of wood particle/PP/coupling agent composites. The goal of this study was to investigate the effects

of OBC-toughened PP blends prepared with various concentrations of MAH on the fracture behavior and mechanical properties of the resulting wood-based composites.

2. Experimental

2.1. Materials

Polypropylene (PP, T30S) with a melt flow index (MFI) of 3.8 g/10 min (230 °C, 2.16 kg) was purchased from Daqing Petrochemical Company (Daqing, China). Olefin block copolymer (OBC, 9010) and octene-ethylene copolymer (POE, 8210) were supplied by the Dow Chemical Company. Maleic anhydride (MAH), dicumyl peroxides (DCP), and xylenes were provided by Tianjin Bochen Company (Tianjin, China). All chemicals and solvents were reagent-grade products and were used without further purification. Poplar (*Populus alba*) wood particles (WPs) were supplied by Harbin Yongxu Company (Harbin, China), and were passed through a 40 mesh (425 µm) sieve before use.

2.2. Fabrication of the modified matrix and their composites

Grafting of MAH onto the polymer matrix was performed using a co-rotating parallel twin-screw extruder (L/D ratio of 56, AK53, Nanjing Rubber and Plastics Machinery Co., Ltd., Nanjing, China). The MAH and DCP were dissolved in acetone, and then sprayed onto the polymer matrix before being fed into the extruder. The rotary speed of the screws was set as 300 rpm (residence time of 90 s), using a feeding rate of 36 kg h⁻¹. The temperature profile of the barrel from the hopper to the die was set as follows: 140, 150, 170, 180, 190, 190, 190, 190, 190, 200, 195, 180, and 170 °C. The extrudates were pelletized and injection-molded (MA900IIS, Haitian Plastic Machinery Group Co., Ltd., Ningpo, China) into standard specimens for mechanical testing. The injection and mold temperatures were 200 and 25 °C, respectively. The concentrations of DCP and MAH were calculated relative to the mass of the whole matrix (Table 1).

Wood particles (WPs) were dried in an oven at 103 °C for 24 h and subsequently mixed with polymer blends at a weight ratio of 6:4 in a heated intensive mixer for 10 min (Table 2). The mixtures were compounded using a co-rotating parallel twin-screw extruder (LD/ratio of 36, SJH30 Nanjing Rubber-Plastic Machine Ltd., Nanjing, China) with a speed of 50 rpm. The extrusion temperature ranged from 160, 170, 180, 180, 180, 170 to 160 °C over seven zones along the extruder barrel. The resulting blend was pelletized. The pellets were extruded into a profile measuring 100 mm × 4 mm (width × thickness) using a single-screw extruder system (L/D ratio of 8, SJ45, Nanjing Rubber-Plastic Machine Ltd., Nanjing, China). The extrusion temperature was set as follows: 145 °C in the melting zone, 155–180 °C in the pumping zone, and 177 °C in the die zone.

2.3. Characterization

2.3.1. Fourier-transform infrared spectroscopy

The polymer matrix was dissolved in refluxing xylene at 140 °C for 1 h and then precipitated in excess acetone [22]. The purified samples

Table 1
Formulations of the polymer blends.

Samples	PP (wt%)	OBC (wt%)	DCP (wt%)	MAH (wt%)
Virgin PP	100	–	–	–
PO	90	10	–	–
PM0.4	100	–	0.9	0.4
POM0.2	90	10	0.9	0.2
POM0.4	90	10	0.9	0.4
POM0.6	90	10	0.9	0.6
POM0.8	90	10	0.9	0.8
PPOM0.6	90	10 (POE)	0.9	0.6

Table 2
Formulations of the composites in extruding.

Samples	WPs (wt%)	Virgin PP (wt%)	MAH-PP (wt%)	MAH-PP/OBC (wt%)
WP	60	40	–	–
WPO	60	–	–	40 (PO)
WPM0.4	60	–	40	–
WPOM0.2	60	–	–	40
WPOM0.4	60	–	–	40
WPOM0.6	60	–	–	40
WPOM0.8	60	–	–	40
WPPOM0.6	60	–	–	40 (PPOM0.6)

were hot-pressed into thin films at 190 °C and analyzed using a spectrometer (Nicolet 6700, Nicolet Company, Madison, Wisconsin, USA). The measurements were performed using an attenuated total reflection (ATR) technique. The data were collected from 4000 cm^{-1} to 400 cm^{-1} at a resolution of 4 cm^{-1} with 32 scans. The signals were collected using a liquid nitrogen-cooled MCT-A detector.

2.3.2. Mechanical testing

Type-I dumbbell-shaped tensile specimens measuring $165 \times 13 \times 4 \text{ mm}^3$ were tested using a universal testing machine (Model RGT-20A, Shenzhen, China) with a gauge length of 50 mm. The tests were conducted at a crosshead speed of 50 mm min^{-1} for polymer matrix (ASTM D638-10) and 5 mm min^{-1} for composites (ASTM D638-10), respectively.

Specimens measuring $80 \times 12.7 \times 4 \text{ mm}$ were used for the flexural test of the matrix and composites using the same universal mechanical machine according to the specification of ASTM D790-10. A crosshead speed of 2 mm min^{-1} and span length of 64 mm were used for the test.

The notched Izod impact samples with “V”-type notch depth of 2.7 mm of polymer matrix (ASTM D256-10) and unnotched Izod impact samples of composites (ASTM D4812-06) were tested by an impact tester (XJ-50G, Chengde, China). The sample sizes of the matrix and composites were $80 \times 12.7 \times 4 \text{ mm}$ and $80 \times 10 \times 4 \text{ mm}$, respectively. Eight replicates were used for each test.

2.3.3. Water uptake and dimensional swelling determination analysis

Specimens measuring $25 \times 25 \times 4 \text{ mm}$ were oven-dried at 50 °C for 24 h and weighed. These specimens were subsequently immersed in distilled water at $23 \pm 1 \text{ °C}$ for 7 weeks. At the stated periods, the specimens were removed from the water bath and the surface water was wiped off with a paper towel to measure the weight and thickness. The water uptake and dimensional swelling of each specimen were calculated based on dry weight and dry dimension, respectively. Five replicates were used for each specimen.

2.3.4. Dynamic mechanical analysis

Specimens $35 \times 12 \times 4 \text{ mm}$ in size were measured in a dynamic mechanical analyzer (Q800, TA Instruments Inc., New Castle, DE, USA). Tests were performed using a three-point flexural mode with oscillating amplitude of 50 μm and a frequency of 1 Hz. The temperature was swept from -20 to 130 °C at 5 °C min^{-1} .

2.3.5. Morphological analysis

The impact specimens were frozen in liquid nitrogen for 10 min and immediately broken perpendicular to the extrusion direction. The fractured surfaces were subsequently dried, sputter-coated with gold, and observed using a scanning electron microscope (SEM, Quanta 200, FEI Company, Hillsboro, OR, USA) at an acceleration voltage of 12.5 kV.

3. Results and discussion

3.1. Grafting modification of PP and PP/OBC blends

Compared with the pattern for virgin PP, new absorption bands at 721 cm^{-1} and 1785 cm^{-1} were observed in all MAH-grafted PP/OBC blends, which can be assigned to absorption bands of hydrocarbon ($-\text{C}-\text{H}-$) in-plane oscillation of the OBC skeleton and carbonyl ($-\text{C}=\text{O}$) stretching, respectively (Fig. 1a). The relative graft degree of MAH grafting onto polymer blends was measured as the absorption ratio (R , in %) of the band at 1785 cm^{-1} to the band at 2722 cm^{-1} (characteristic absorption bands of the PP backbone) [22]. The relative grafting degree of MAH increased with increasing MAH concentration (Fig. 1b). Notably, the MAH graft degree of POM0.4 was higher than that of MAH-grafted PP (PM0.4) at the same MAH concentration. This result may be attributed to the higher grafting reactivity of OBC compared to that of the PP matrix [7].

As shown in Fig. 2a and b, PM0.4 showed a rougher fracture surface than virgin PP. This may be due to the initiator-induced degradation of PP chain, thereby making the PM0.4 more brittle [18]. With the addition of OBC (Fig. 2c), many voids were observed in the fracture surface due to the debonding of OBC particle, indicating that PP is immiscible with OBC. With further grafting of PP/OBC blends with MAH (Fig. 2d), fewer voids were observed on the fracture surface compared to the unmodified blends. This indicates that grafting modification with MAH can improve interfacial compatibility between PP and OBC.

3.2. Mechanical properties of the matrix

Apart from the target grafting reaction, other reactions such as PP-chain scission and branching reactions between PP and OBC have been observed [23]. PM0.4 showed a low value of elongation at break (43%), while a typical ductile plastic behavior with large strain was observed for virgin PP and the samples containing OBC (Fig. 3a). This brittle behavior of PM0.4 can be attributed to the peroxide-induced chain scission reaction, where these short chains increased the molecular slippage to result in fracture with low strain value. The tensile strain of PO and POM0.4 were increased by 192 and 152%, respectively, compared to virgin PP. However, the slope of the tensile stress-strain curve decreased with the addition of low-modulus OBC (Fig. 3a). Grafting of PP/OBC blends with MAH slightly increased the slope due to the improvement of the interfacial compatibility between PP and OBC.

The impact strength of PM0.4 (6.3 kJ m^{-2}) decreased by 63.0%, compared to that of virgin PP (17.0 kJ m^{-2}) (Fig. 3b). The tensile and flexural strength of PM0.4 were 34.8 MPa and 47.5 MPa, respectively (Fig. 3c), which slightly decreased compared to virgin PP. However, the tensile and flexural modulus values of PM0.4 remained equal to that of virgin PP (Fig. 3d). These results can be explained as follows: With application of the tensile and flexural stress, these short-chain molecules had sufficient time to relax and dissipate the stress, but high-strain rate occurred under impact loading, resulting in ultimate failure with low-energy dissipation. With the addition of 10 wt% of OBC, the impact strength of POM0.4 increased to 28.4 kJ m^{-2} (Fig. 3b) with 366% and 67% improvements compared to that of PM0.4 and virgin PP, respectively, demonstrating the significant toughening ability of OBC. Due to the low modulus of OBC, the tensile strength (28.54 MPa) and modulus (0.54 GPa) of POM0.4 were decreased by 18 and 19%, respectively, compared to PM0.4 (Fig. 3c and d). The flexural strength and modulus values were similarly decreased. Additionally, grafting of the PP/OBC blends with MAH decreased the impact strength due to the chain scission of the PP chain. However, by comparing the toughening effect of commercial elastomer-POE (as shown in Fig. 1S), a significant improvement in impact resistance and a slight decrease in stiffness were observed in the OBC-toughened PP blends, suggesting the superior toughening ability of OBC.

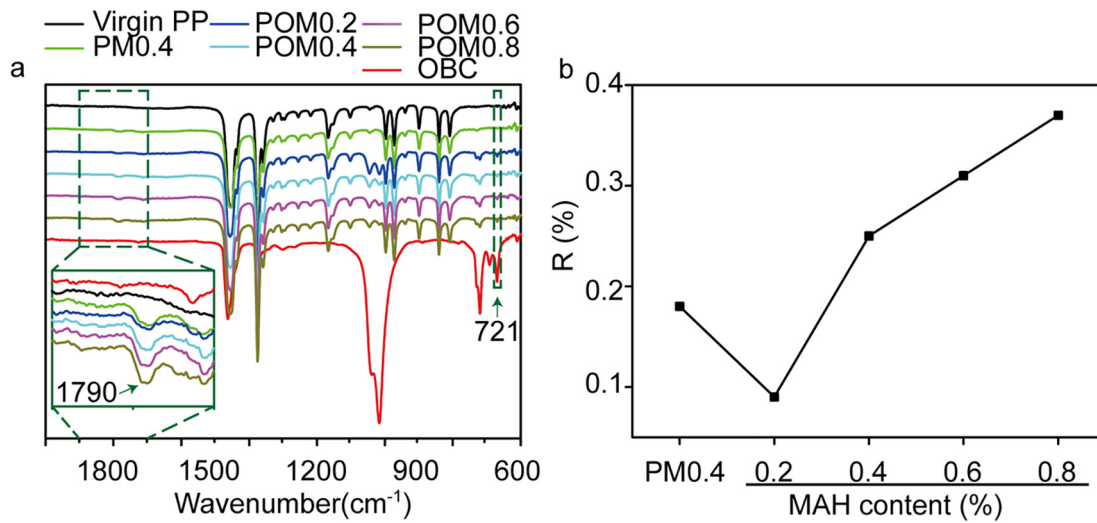


Fig. 1. (a) FTIR spectra of virgin PP, OBC, grafted PP, and grafted PP/OBC blends with various MAH concentrations and (b) MAH absorption ratio (R) of the grafted PP and the grafted PP/OBC blends at various MAH content.

3.3. Mechanical properties of the composites

For composites prepared with the unmodified PP/OBC blends as the matrix, the tensile and flexural strengths were decreased by 16.1% and 22.9%, respectively, and the tensile and flexural modulus were decreased by 12.1% and 14.1%, respectively, compared to the virgin PP/wood composites (WP) (Fig. 4). The impact strength of WPO and WP exhibited no obvious change. This can be attributed to the adverse softening effect of the low-modulus OBC and the poor compatibility of OBC with wood particles and PP.

Compared with WP, the MAH-grafted PP matrix improved the tensile properties, flexural properties, and impact strength, but had less effect on the flexural modulus (Fig. 4). For materials at the same MAH concentration, the blending of OBC with PP caused a decrease in the strength and modulus of the PP/OBC blends (Fig. 3a and b), while the strength and modulus in the wood particle-filled composites were well maintained (Fig. 4a and c). In addition, the impact strength of WPOM0.4 was increased by 38% relative to that of WPM0.4 (Fig. 4a). This result indicates that the toughening ability of OBC was mainly dependent on the degree of interfacial adhesion between the components.

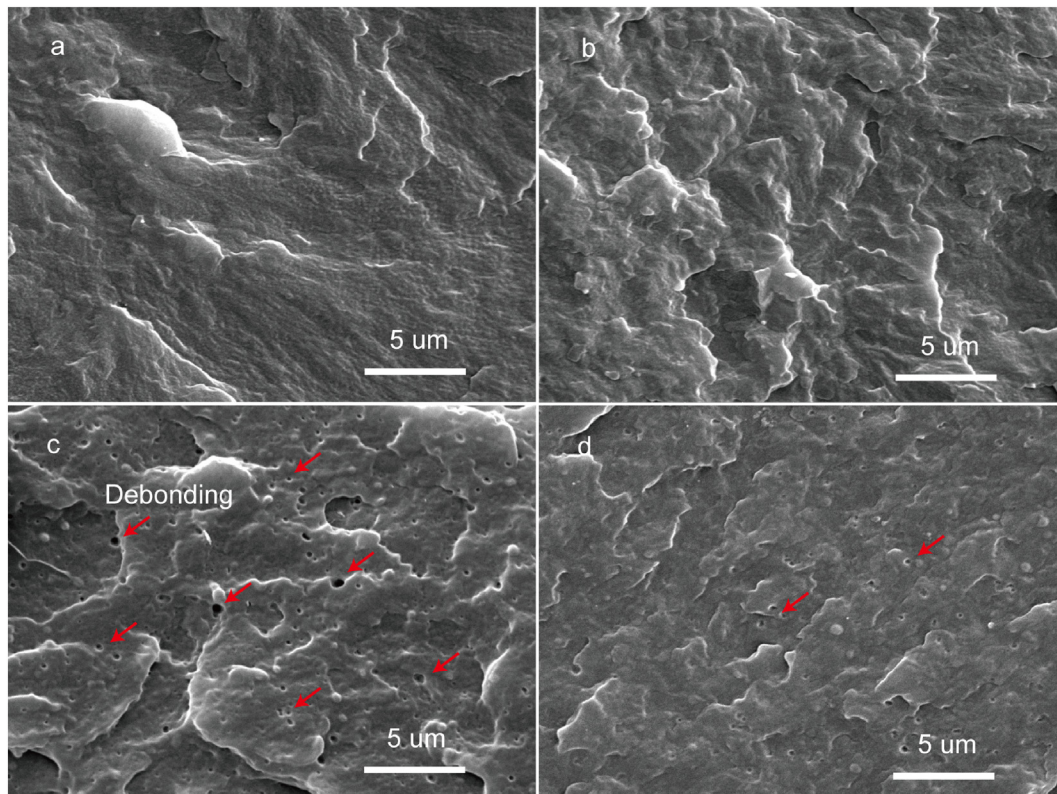


Fig. 2. Micrographs of fracture surface for virgin PP (a), PM0.4 (c), PO (b), POM0.4 (d).

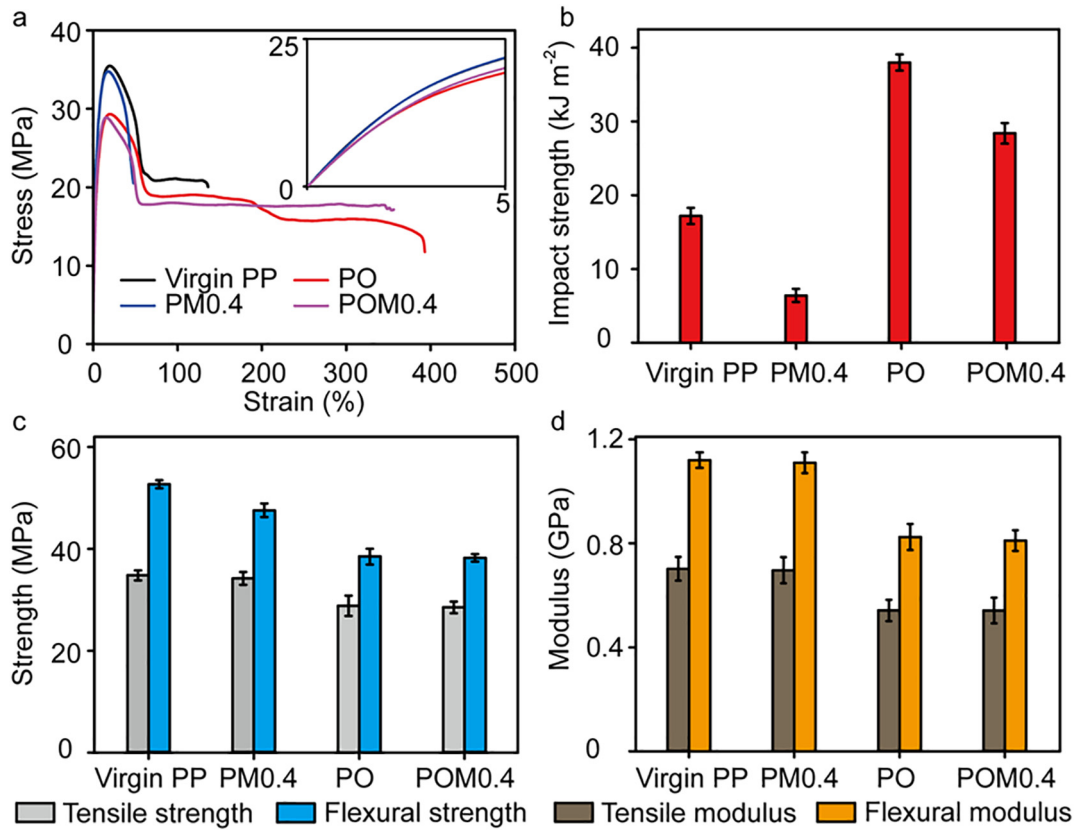


Fig. 3. (a) Stress-strain curves, (b) impact strength, (c) tensile and flexural strength, (d) tensile and flexural modulus of the matrix.

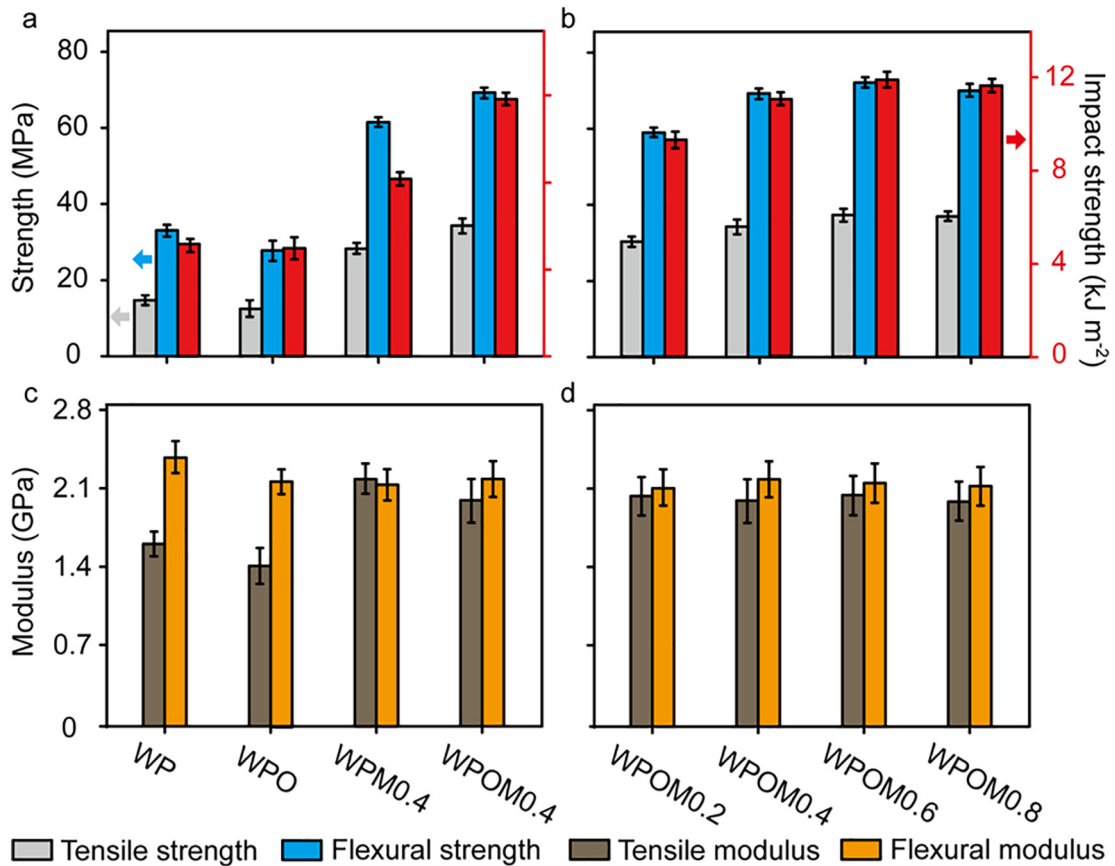


Fig. 4. (a and b) Tensile, flexural, and impact strength and (c and d) tensile and flexural modulus of the composites.

As shown in Fig. 5a, the fracture surface of WP showed the pull-out of wood particles, resulting in the formation of many large voids. The wood particle surfaces were smooth and clean, lacking the adhering matrix, thus indicating poor interaction between wood particles and the PP matrix. By contrast, both of the composites based on PM0.4 and POM0.4 possessed wood particles that were embedded in the modified matrix. The interfacial boundary became less distinct, and some polymer residuals were apparent on the surface of the wood (Fig. 5b and c). Therefore, the poor mechanical properties of WP and WPO can be attributed to weak interfacial bonding between wood particles and the PP matrix, as evidenced by the SEM micrographs (Fig. 5a). There were enhanced interphase interactions between PP/OBC blends and wood particles as a result of the matrix modification with MAH. Additionally, the grafted blends with short PP chains showed excellent wettability to the hydrophilic wood particles, thereby improving the particle's dispersibility [7]. When these composites were subjected to loading, stress was sufficiently transferred from the modified matrix to the particles through effective interfacial shear stress, resulting in improved strength and modulus [5]. However, in the elastomer-toughened systems, the good compatibility among PP, OBC and wood particles was able to provide sufficient adhesion to initiate and/or promote local matrix plastic deformation, resulting in a high impact resistance of the resulting composites [18]. In addition, the MAH-grafted OBC may encapsulate the wood particles to create embedded structure in the composites, which may also contribute to the improvement of the modulus. Finally, a better strength-toughness balance was also observed in OBC toughened composites compared to the POE toughened composites (as shown in Fig. S2), which further confirms the good toughening effect of OBC.

The tensile and flexural strength showed an increasing trend as the MAH content increased to 0.6 wt% (Fig. 4b). Compared with WPOM0.4, the tensile strength, flexural strength, and impact strength of WPOM0.6 were increased by 9%, 3%, and 11%, respectively. This result indicated that the grafting of more hydrophilic anhydride groups on the blends improved the stress transfer ability of the materials. For these composites, the stresses were effectively transferred from the modified matrix to the wood particles and exceeded the fracture strength of the wood particles, as evidenced by the SEM micrographs (Fig. 5c). Therefore, the wood particles (60 wt%) primarily bear the load, resulting in ultimate breakage with large energy dissipation. A further increase of the MAH content did not cause a further increase in the mechanical properties. This result indicated a critical value of MAH grafting, beyond which the strength of the composites cannot be further improved by increasing this MAH content. Unlike tensile and flexural strength, the modulus value is less dependent on interfacial interaction between the matrix and filler but more sensitive to large deformations of the material itself [20]. Thus, changes in MAH content had no obvious influence on the modulus values of the resulting composites (Fig. 4d).

The impact strength of the composites also showed an increasing trend with the increase in MAH content (Fig. 4b). This result indicated that stronger interphase interactions can more efficiently constrain crack initiation and then stabilize and premature crack propagation

along the interface. In this system, additional plastic deformation processes can be induced to dissipate a great amount of impact energy. However, the fracture energy may also increase through particle fracture rather than pull-out, thus resulting in high impact toughness of the composites.

3.4. Dimensional stability of the composites

Compared with the unmodified composites (WP), both the water uptake and thickness swelling of WPM0.4 and WPOM0.4 were considerably reduced due to MAH grafting of the matrices (Fig. 6a and c). This result may be mainly due to the increased dispersion of wood in the plastic matrix and the improved interfacial bonding between the wood particles and plastic as a result of the MAH grafting. This improved interfacial adhesion can slow the rate of water diffusion between the wood particles through the interfacial defects. The water uptake and thickness swelling of WPOM0.4 were lower than those of WPM0.4, indicating that WPOM0.4 had stronger interfacial interaction among the components than WPM0.4 due to the higher MAH grafting degree of POM0.4 (Fig. 1b). At the same OBC level, increased MAH content resulted in decreased water absorption and thickness swelling (Fig. 6b and d). This enhanced dimensional stability can be attributed to the greater formation of ester bonds, thereby reducing the water diffusion rate and limiting swelling.

3.5. Dynamic mechanical properties of the composites

Compared with WP, the storage modulus values of WPM0.4 and WPOM0.4 were remarkably increased due to the improved interfacial adhesion (Fig. 7a). This was consistent with the measured mechanical properties and SEM observation of these composites. Usually, enhanced interfacial adhesion improves stress transfer from the matrix to the particles, thereby increasing the storage modulus of the resulting composites. However, the modified composites containing OBC exhibited a significant decrease in the storage modulus compared to the storage modulus of WPM0.4. This decrease was not consistent with the measured static mechanical properties (Fig. 4). This suggests that although the interfacial adhesion of WPOM0.4 was better than that of WPM0.4 (Fig. 6), the softening effect of the low-modulus OBC exceeded the strengthening effect due to the enhanced interfacial bonding. With further increase of the MAH content, the storage modulus increased (Fig. 7b), suggesting more efficient transfer of the stress from the matrix to the wood particles through more effective interface interactions [24].

The damping ratio ($\tan \delta$) is dependent on the nature of individual components and the degree of interfacial interaction [25]. The $\tan \delta$ values of WPM0.4 and WPOM0.4 were smaller over the whole temperature range than those of WP (Fig. 7c). Given the above-described mechanical properties and dimensional stability, the improved compatibility between the modified matrix and the wood particles reduced the interphase damping. The grafted matrix surrounding the wood surface hindered the molecular motion, and then decreased the energy dissipation during the relaxation process. In the low

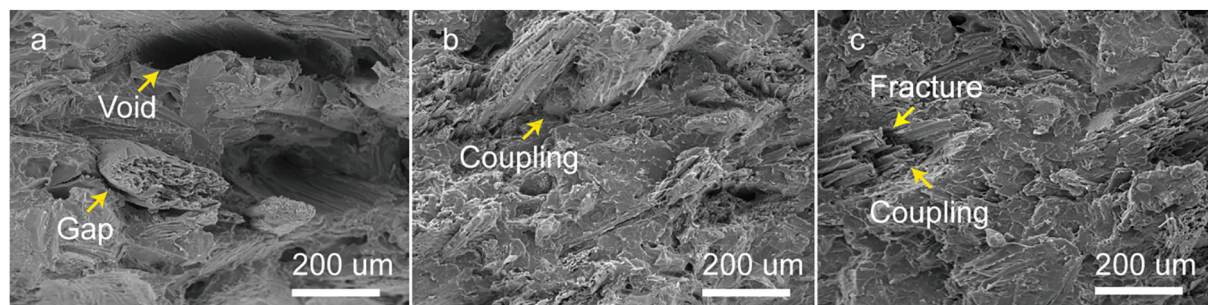


Fig. 5. Micrographs of fracture surface for (a) WP, (b) WPM0.4, and (c) WPOM0.4.

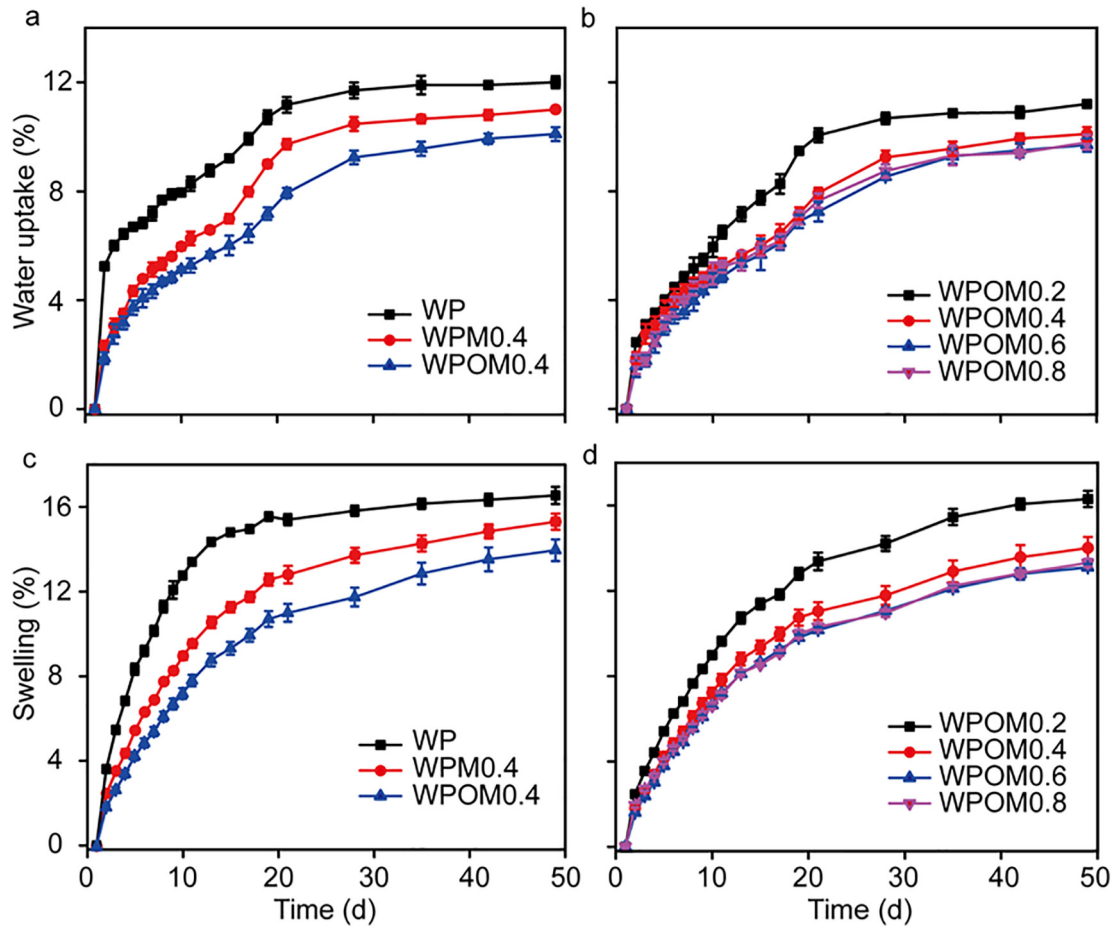


Fig. 6. (a and b) Water uptake and (c and d) dimensionally swelling of the unmodified and modified composites.

temperature range, it was noted that the WPOM0.4 showed a lower $\tan \delta$ value due to the strong interfacial adhesion. With increasing temperature, WPOM0.4 showed a higher $\tan \delta$ value at temperatures above $\sim 80^\circ\text{C}$ than WPM0.4. A slight peak was also observed above 60°C , which represented the α -relaxation temperature for the PP chain at which chain rotation of the polymer occurs [19,25]. In the absence of OBC, there was a relatively rigid interfacial layer around the particles in WPM0.4, which allowed the rigid wood particles to bear a high amount of stress with less deformation, thereby resulting in low energy dissipation at the interface. However, the separate low-modulus OBC in the composites caused more interface areas and defects, allowing the dissipation of more viscous energy in the expanded interface. Therefore, with increasing temperature, the chain rotation of the low-modulus OBC may result in deformation and friction at the interface, increasing the viscous energy dissipation and corresponding to high values of $\tan \delta$.

With increasing MAH content, the $\tan \delta$ values of the composites exhibited a decreasing trend (Fig. 7d). This result suggests that a higher MAH concentration allowed the formation of more ester bonds between the matrix and the wood particles, resulting in less viscous energy dissipation in the interface. As the temperature increased, the slope of the $\tan \delta$ curve decreased with increasing MAH content. This suggests that the better interfacial adhesion between the components contributed to an enhanced ability to restrict relaxation.

3.6. Analysis of deformation and failure mechanism

The deformation and failure process of wood-filled PP composites can be summarized as an energy dissipation mechanism, including matrix-related (crack initiation and propagation) and particle-related (pull-out, debonding, and fracture) mechanisms [21]. Sudar et al.

found that the extent of plastic deformation of polymer matrix largely determined the impact resistance of wood-filled composites and the stiffness was well maintained due to adequate particle embedding [15,16].

For unmodified composites WP and WPO, the higher content (60 wt %) wood particles could not be evenly dispersed in the matrix, resulting in the agglomeration of the particles (Fig. 8). When external stress load is applied, the wood particles acted as stress concentrators to promote crack initiation and propagation along the weak interfacial adhesion in the composites, resulting in failure of composites with low energy dissipation. A previous study [23] of the composites with MAH-grafted polymer as matrix reported sufficient interfacial bonding and even dispersion of particles in the matrix. In this study, there was significant variation in the impact resistance of modified composites prepared with and without OBC (Fig. 4c), suggesting that there may be differences in the deformation and failure mechanisms.

For WPOM0.4, both a relatively rigid grafted-PP interfacial layer and an elastomeric grafted-OBC interfacial layer were expected to form around the wood particles [19]. However, only the rigid interface layer formed in WPM0.4. The lower impact resistance of WPM0.4 implied that crack initiation was limited by the improved interfacial bonding, which also restricted the release of the constraint imposed by the interfacial adhesion and rigid particles. Crack propagation would cause the breakage of wood particles with less plastic deformation. The fracture toughness of WPM0.4 was mainly due to wood particle-related mechanisms. When the OBC-strengthened composites were subjected to impact loading, there was initial internal cavitation of low-modulus elastomer that subsequently gave rise to the formation of micro-voids. The strong interfacial adhesion and relaxation of elastomeric interface would stabilize the initiation of deformation at initial impact. These

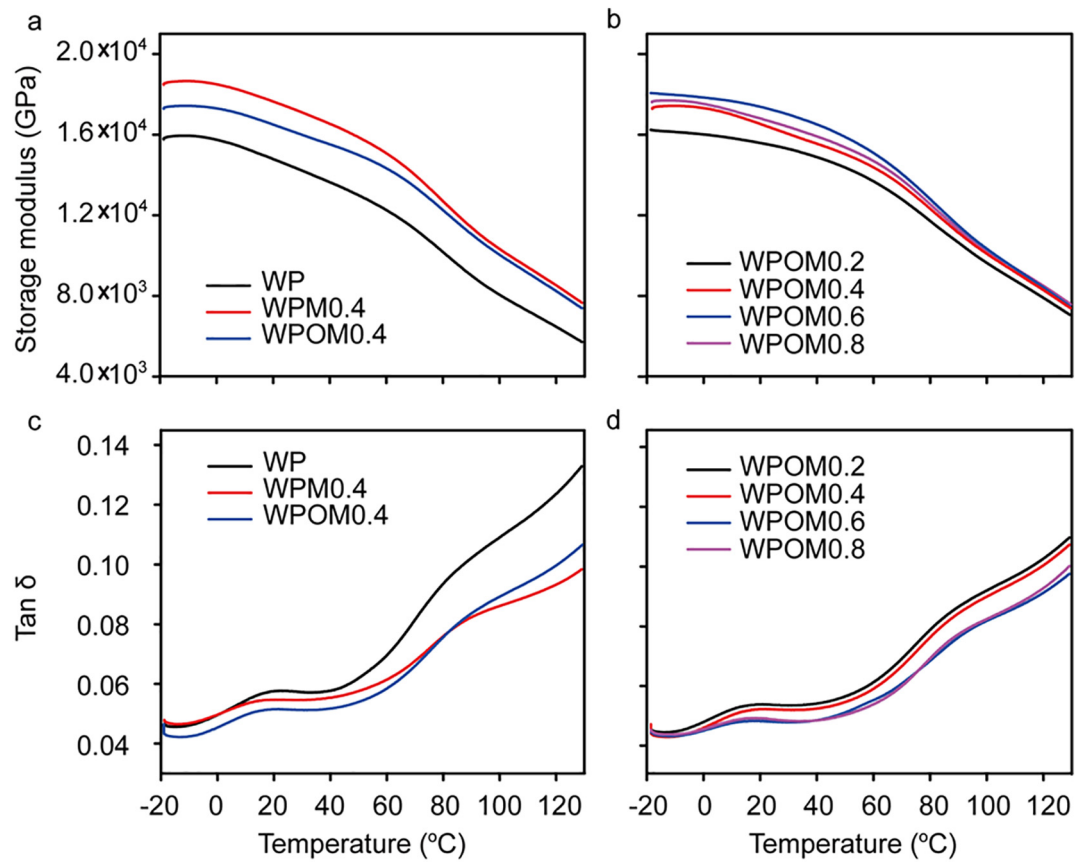


Fig. 7. (a and b) Storage modulus and (b and d) damping ratio ($\tan \delta$) of the unmodified and modified composites.

local deformation processes would further release the imposed plastic constraint around wood particles, thereby inducing more plastic deformation in the matrix. In addition, the separate dispersion of non-grafted

OBC in the PP matrix also induced some plastic deformation by debonding around the particles or the PP matrix. A premature crack could be stably propagated due to good three-phase interfacial bonding,

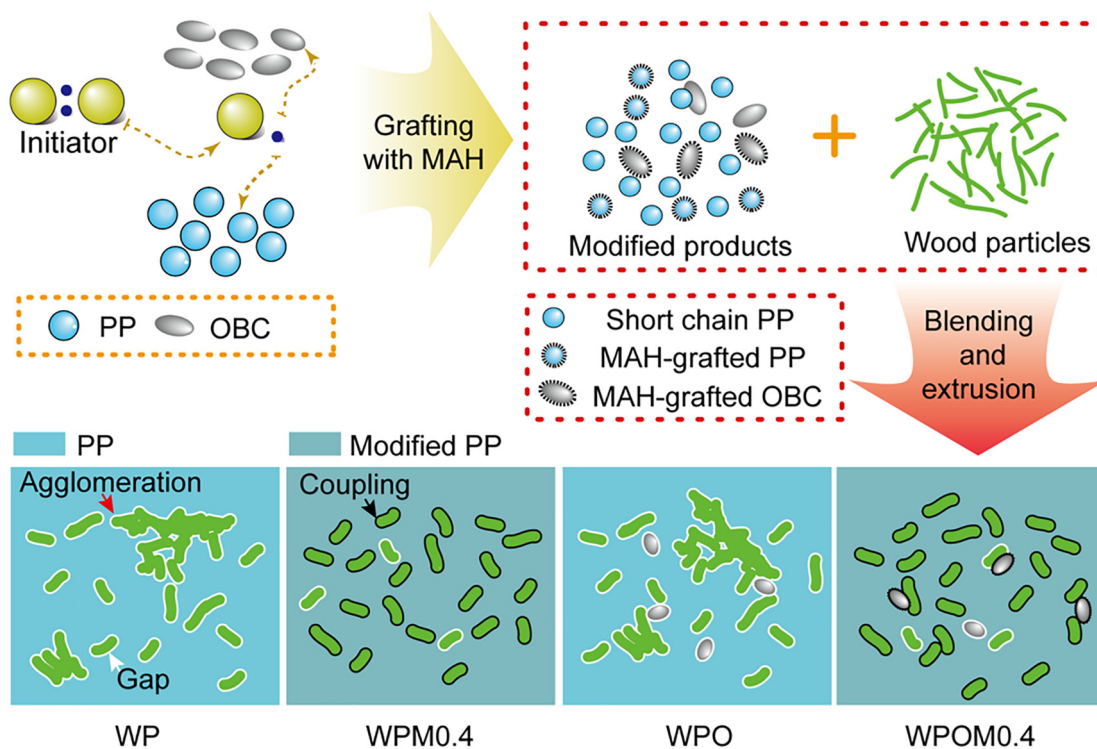


Fig. 8. Schematic representations of melt grafting process and the distribution of the components in WP, WPM0.4, and WPOM0.4.

resulting in ultimate failure with significant energy dissipation. Therefore, the fracture toughness of WPOM0.4 was dominated by plastic deformation and the fracture of particles. In contrast, in the absence of adequate interfacial bonding between the component, a large local stress would form due to particle agglomeration, which would further promote cavitation and hinder the plastic deformation processes [24]. In this case, an unstable crack would rapidly propagate, leading to ultimate brittle failure. This provides a possible explanation for the observed synergetic toughening effect of modification of the PP matrix with both OBC and MAH on the impact strength of high wood particle-filled composites.

4. Conclusions

In this study, wood particle-reinforced PP composites with high impact toughness and stiffness were successfully fabricated via blending OBC and grafting the matrix with MAH. The results confirmed that the toughening effect of OBC on wood particle-filled PP composites was better than that of the commonly used commercial elastomer (POE). The grafting modification strategy utilized here can be used to effectively improve the interfacial compatibility of multiphase systems. Based on the deformation and failure mechanism, the sufficient interfacial interactions and the existence of an elastomeric interface promoted the initiation of massive local plastic deformation, thereby improving impact strength. This simple, economical, and environmentally friendly manufacturing process results in material with excellent toughness-stiffness balance, making this material attractive for many applications.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matdes.2019.107918>.

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